Cultural Bias Matters: A Cross-Cultural Benchmark Dataset and Sentiment-Enriched Model for Understanding Multimodal Metaphors

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Abstract

Multimodal metaphor understanding is a core challenge in natural language processing, yet existing datasets and methods are predominantly English-centric, raising concerns about cultural biases and overstated generalization. We introduce **MultiMM**, the first cross-cultural multimodal metaphor benchmark comprising 8,461 text-image advertisement pairs spanning Chinese (CN) and English (EN), annotated for metaphor occurrence, source/target domains, and sentiment polarity. Building on MultiMM, we conduct a rigorous evaluation of 18 baselines across textual, visual, and multimodal paradigms—including large multimodal language models (MLLMs)—and reveal systematic performance gaps between languages that expose cultural bias. To address this gap, we propose **SEMD** (Sentiment-Enriched Metaphor Detection), a multimodal fusion model that integrates visual features from ViT, multilingual text features from mBERT, and auxiliary sentiment embeddings via concatenation and a feed-forward classifier. SEMD achieves state-of-the-art performance on metaphor detection (80.16% EN, 77.79% CN F1) and sentiment analysis (75.69% EN, 70.51% CN F1), consistently outperforming strong multimodal baselines. Cross-cultural analyses and translation-based experiments further quantify the cultural transfer gap and demonstrate that affective signals provide a universal bridge for robust multimodal metaphor understanding.

1 Introduction

Metaphor is a pervasive and indispensable feature of human language and cognition (Lakoff, 1994; Kövecses, 2010; Ortony et al., 2022). Far from being a mere stylistic device, metaphors structure how we conceptualize abstract domains in terms of more concrete ones, and are deeply embedded in everyday communication (Tong et al., 2021). With the proliferation of multimodal content—images paired with text—metaphorical meaning increasingly spans both visual and linguistic channels (Forceville and Urios-Aparisi, 2009; Forceville, 2002). Advertising is a particularly rich domain: ads routinely exploit visual metaphors to convey persuasive messages with emotional resonance (Phillips and McQuarrie, 2004; Ye et al., 2021).

Automatic metaphor processing has advanced substantially in recent years, driven by deep learning models and benchmark datasets (Shutova et al., 2010; Zhang et al., 2021; Xu et al., 2022; Tong et al., 2024a). However, the field suffers from two interrelated limitations. First, most work focuses exclusively on English, with limited attention to other languages, particularly Chinese (Chen et al., 2023; Zhang et al., 2023). Second, multimodal metaphor datasets and models rarely consider cultural variation, despite strong evidence that metaphor conventions differ across cultures (Kövecses, 2008; Ibarretxe-Antuñano, 2013; Forceville, 2017). As a result, models trained or evaluated on English data may exhibit cultural biases that go undetected when no cross-cultural benchmark exists.

This paper addresses both gaps through two main contributions. First, we construct **MultiMM**, a large-scale cross-cultural multimodal metaphor benchmark with 8,461 Chinese and English advertisement text-image pairs, annotated for metaphor occurrence, source and target conceptual domains, and sentiment polarity. MultiMM supports two tasks—metaphor detection and sentiment analysis—and includes a rigorous annotation protocol with multi-annotator adjudication and inter-annotator reliability metrics. To our knowledge, MultiMM is the first benchmark that directly enables cross-cultural comparison of multimodal metaphor understanding.

Second, we propose **SEMD**, a Sentiment-Enriched Metaphor Detection model that fuses visual, textual, and affective signals. Our key in-

sight is that sentiment is an intrinsic property of metaphorical language (Citron and Goldberg, 2014; Mohammad et al., 2016; Stanley et al., 2021) and can serve as a complementary modality to improve metaphor detection across cultural contexts. SEMD integrates ViT image features, mBERT multilingual text features, and auxiliary sentiment embeddings through concatenation and a feed-forward classifier.

We evaluate 18 baselines spanning text-only models (mBERT, XLM-R, LaBSE, GPT-3.5, LLaMA variants, DeepSeek), vision-only models (VGG16, ResNet50, ViT), and multimodal models (CMGCN, CES, MET, MFC, GPT-4o, LLaVA, Gemini, Qwen2.5-VL). Our experiments reveal clear cultural biases: all models perform systematically worse on Chinese than English data, MLLMs underperform specialized baselines, and the cultural transfer gap widens when language translation is used as a proxy for cross-cultural adaptation. SEMD achieves best-in-class performance on both tasks and both languages, and ablation studies confirm the contribution of sentiment enrichment and feature concatenation.

Our contributions are summarized as follows:

- **MultiMM**: the first cross-cultural multimodal metaphor benchmark with 8,461 CN-EN advertisement pairs, labeled for metaphor occurrence, source/target domains, and sentiment.
- A comprehensive evaluation suite covering 18 baselines across modalities, revealing systematic cultural bias in multimodal metaphor processing.
- **SEMD**: a sentiment-enriched multimodal baseline that consistently improves cross-cultural metaphor detection and sentiment analysis over strong multimodal models.
- Empirical analyses including cross-cultural data statistics and translation experiments that quantify cultural transfer gaps and guide culturally robust modeling.

2 Related Work

2.1 Metaphor Processing

Metaphor identification has been studied from both linguistic and computational perspectives (Steen et al., 2010; Shutova et al., 2013, 2017). Early work relied on manually crafted rules and semantic resources (Mason, 2004; Birke and Sarkar, 2006). The advent of neural networks and pre-trained language models spurred significant progress in verb- and token-level metaphor detection (Mao et al., 2018; Choi et al., 2021; Su et al., 2021b; Ge et al., 2022a; Tian et al., 2024).

Interest in the relationship between metaphor and emotion has grown in parallel (Citron and Goldberg, 2014; Mohammad et al., 2016): metaphorical expressions tend to be more emotionally engaging than their literal counterparts, and sentiment signals can aid metaphor identification. Recent work has also explored metaphor generation (Stowe et al., 2021) and the limits of large language models on metaphor understanding tasks (Wachowiak and Gromann, 2023; Tong et al., 2024a,b).

2.2 Multimodal Metaphor

Multimodal metaphors express meaning through the interaction of multiple sign systems, such as text and image (Forceville and Urios-Aparisi, 2009; Forceville, 2021). Forceville's foundational work on pictorial and multimodal metaphors in advertising (Forceville, 2002, 2017) established a theoretical basis for this domain. Computational approaches have since developed datasets and models for multimodal metaphor recognition (Zhang et al., 2021; Xu et al., 2022; Su et al., 2021a; Lu et al., 2025; Zhang et al., 2025; Ge et al., 2022b).

MultiMET (Zhang et al., 2021) introduced a dataset of Chinese text-image pairs for multimodal metaphor understanding, while MET-Meme (Xu et al., 2022) focused on memes in English with metaphor and sentiment annotations. MultiCMET (Zhang et al., 2023) proposed a Chinese benchmark for multimodal metaphor. MetaCLUE (Akula et al., 2023) offered a broader suite of visual metaphor tasks. Unlike these prior efforts, MultiMM explicitly pairs Chinese and English data to enable direct cross-cultural comparison.

2.3 Cultural Aspects of Metaphor

There is substantial evidence that metaphor use varies across cultures (Kövecses, 2008; Ibarretxe-Antuñano, 2013; Pritzker, 2003). While some conceptual metaphors appear universal (e.g., TIME IS MONEY), the specific source domains, emotional associations, and pragmatic conventions differ markedly between Eastern and Western cultures (Chin et al., 2021; Piekkari et al., 2020; Dwairy, 2009). However, cross-cultural computational metaphor processing remains understudied. Our work directly targets this gap by constructing

a balanced bilingual benchmark and systematically measuring cultural bias across model classes. Computational studies of cross-lingual sentiment (Mohammad et al., 2016) have similarly identified language-contingent affect mappings, suggesting that metaphor and sentiment share culturally-bound properties that motivate their joint modeling.

2.4 Multimodal Learning

Our work builds on advances in multimodal representation learning. Pre-trained vision encoders such as VGG (Simonyan and Zisserman, 2014), ResNet (He et al., 2016), and ViT (Dosovitskiy et al., 2020) have become standard for image feature extraction. Multilingual pre-trained language models, including mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), and LaBSE (Conneau et al., 2020), support cross-lingual transfer. Multimodal models for sentiment and figurative language analysis (Liang et al., 2022; Blaier et al., 2021; Maity et al., 2022; Zhang et al., 2024) provide relevant architectural patterns, while large multimodal models (Achiam et al., 2023; Liu et al., 2024; Team et al., 2023; Bai et al., 2025) increasingly serve as zero-shot baselines.

3 The MultiMM Dataset

3.1 Overview

MultiMM is a cross-cultural multimodal metaphor benchmark consisting of 8,461 text-image advertisement pairs: 4,397 in Chinese (CN) and 4,064 in English (EN). Table 1 summarizes the dataset statistics. Each sample comprises an advertisement image and its associated text (extracted via OCR for CN, originally provided for EN), annotated with four labels: **metaphor occurrence** (metaphorical/literal), **target domain** (the concept being described), **source domain** (the concept used to describe it), and **sentiment category** (negative/neutral/positive).

MultiMM supports two evaluation tasks:

- **Metaphor Detection**: binary classification of whether an ad contains a multimodal metaphor.
- **Sentiment Analysis**: three-way classification of the advertisement's sentiment polarity.

Data is split into training (6,768), validation (846), and test (847) sets.

Item	CN	EN	Total
Total	4,397	4,064	8,461
Metaphorical	2,583	2,189	4,772
Literal	1,814	1,875	3,689
Total Words	145,312	68,189	213,501
Average Words	33	15	24
Training Set Size	3,517	3,251	6,768
Validation Set Size	440	406	846
Test Set Size	440	407	847

Table 1: Statistical overview of MultiMM. CN refers to Chinese data, while EN refers to English data.

3.2 Data Sources and Collection

Chinese Data. Chinese advertisements were collected through two channels: (1) systematic keyword-based Baidu image search spanning product categories (e.g., food, electronics, cosmetics), public service announcements (PSAs), and metaphor-relevant conceptual domains; and (2) a proprietary iFlytek advertisement dataset providing additional diversity.

English Data. English advertisements were sourced from a publicly available dataset covering commercial product ads and PSAs, providing a complementary cultural perspective.

Preprocessing and Filtering. Raw images underwent MD5 hash-based deduplication (Rivest, 1992) to remove exact duplicates. Images were further filtered by excluding non-advertisements, blurry or low-resolution images (minimum 350×350 pixels), and samples with unreadable text. For Chinese ads, text was extracted using an OCR pipeline with manual correction. This pipeline yielded 4,397 CN and 4,064 EN samples after quality filtering.

3.3 Annotation Protocol

Annotation Scheme. Each sample is annotated with four fields: (1) *Occurrence*: whether the ad contains a multimodal metaphor (binary); (2) *Target domain*: the conceptual domain being described in the metaphor (e.g., a product's quality); (3) *Source domain*: the conceptual domain used as the vehicle (e.g., nature, animals); (4) *Sentiment*: the overall sentiment conveyed by the ad (negative, neutral, or positive). Visual source and target domains were verbalized (converted to text descriptions) to enable uniform annotation. Annotation guidelines were derived from the multimodal metaphor literature (Forceville and Urios-Aparisi, 2009; Šorm and Steen, 2018; Steen et al., 2010).

Figure 1: An example advertisement with full MultiMM annotations. The ad contains a visual metaphor where a product's attribute (target domain) is mapped onto a source domain from nature, with overall positive sentiment.

Annotation Teams. Chinese data was annotated by five expert annotators organized in groups of 2+2+1 (two independent annotators, a second pair for cross-check, and an adjudicator). English data was annotated by three annotators in a 2+1 configuration. All annotators completed training rounds with iterative guideline refinement. Multicultural annotator recruitment was employed to mitigate cultural bias in label assignment.

Quality Control. Inter-annotator reliability was assessed using Fleiss' κ (Fleiss, 1971). Results indicate substantial agreement: $\kappa = 0.73$ for metaphor occurrence, $\kappa = 0.70$ for target domain, $\kappa = 0.66$ for source domain, and $\kappa = 0.82$ for sentiment category. An annotation example is shown in Figure 1.

4 Data Analysis

4.1 Cross-Cultural Metaphor Examples

Figure 2 illustrates representative metaphors from Chinese and English advertisements. These examples highlight how cultural norms shape the choice of source and target domains: Chinese ads frequently employ traditional symbols and nature-based imagery, while English ads tend toward direct product comparisons and idiomatic visual tropes.

4.2 Sentiment Distribution

Figure 3 shows the sentiment category distribution across metaphorical and literal samples in CN and EN data. A striking cultural divergence emerges: positive sentiment dominates metaphorical advertisements far more strongly in English (74.75%) than in Chinese (56.42%). This pattern reflects broader cultural differences in advertising conventions, where Western ads tend toward overt positivity while Chinese ads frequently employ neutral or problem-framing strategies. Negative and neutral sentiments are proportionally more common among literal ads in both languages.

(a) Chinese metaphor example

(b) English metaphor example

Figure 2: Examples of multimodal metaphors across cultures. Chinese and English ads exhibit distinct metaphor conventions, reflecting cultural differences in symbolic associations.

4.3 Source Domain Distribution

Figure 4 presents the distribution of source domain vocabulary in Chinese and English advertisements. The source domains differ substantially across the two languages, reflecting culture-specific conceptual mappings. Chinese advertisements frequently draw upon natural imagery and traditional cultural motifs, while English advertisements more commonly employ comparisons to familiar everyday objects or universal abstract concepts. This divergence in source domain vocabulary has direct implications for model design: culturally-specific symbol systems mean that visual representations pre-trained on Western-centric image corpora may underrepresent the visual vocabulary needed for Chinese metaphor interpretation, motivating the use of multilingual and culturally-diverse pre-training data.

4.4 Word Cloud Analysis

To further examine the relationship between source domains and sentiment, Figure 5 shows word

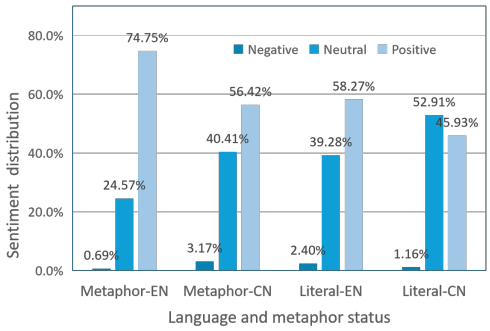

Figure 3: Sentiment category distribution across metaphorical and literal advertisements in Chinese (CN) and English (EN). Positive sentiment is more dominant in EN metaphorical ads (74.75%) than CN (56.42%), revealing cultural divergence in affective framing.

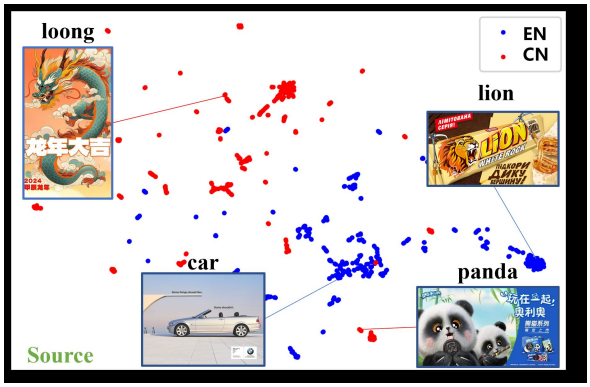

Figure 4: Distribution of source domain vocabulary in Chinese (CN) and English (EN) advertisements. The distinct domain distributions reflect culture-specific conceptual mappings in metaphorical advertising.

clouds of source domain vocabulary stratified by sentiment category in CN and EN. In English, positive-sentiment metaphors cluster around concepts of freshness, brightness, and natural vitality. In Chinese, the distribution across sentiment categories is more balanced, with traditional cultural symbols appearing prominently regardless of sentiment. These differences underscore the importance of culturally-aware annotation and modeling.

5 Methodology

5.1 Task Formulation

Given a text-image pair (T, I), the metaphor detection task requires predicting a binary label $y_m \in \{0, 1\}$ (literal vs. metaphorical), and the sentiment analysis task requires predicting $y_s \in \{-1, 0, 1\}$ (negative, neutral, positive). For metaphor detection, SEMD incorporates an auxiliary sentiment representation; for sentiment analysis, SEMD uses

Figure 5: Word clouds of source vocabulary across sentiment categories in Chinese (CN) and English (EN) advertisements, illustrating culture-specific associations between source domains and affective polarity.

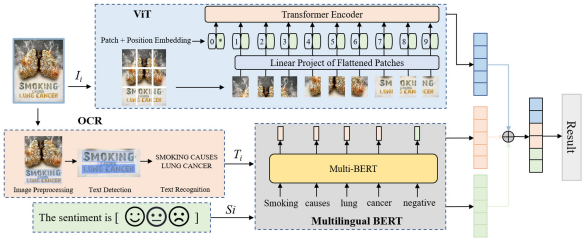

Figure 6: The SEMD architecture. Three branches extract image features (ViT), text features (mBERT from OCR output), and sentiment embeddings (from a sentiment analysis module), which are concatenated and fed into a feed-forward classifier.

image and text features only.

5.2 SEMD Architecture

Figure 6 illustrates the SEMD framework, which consists of three parallel branches:

Image Encoder. We use ViT-base-patch16-224 (Dosovitskiy et al., 2020) to extract visual representations. The input image is divided into 16×16 patches; after self-attention across all patch tokens, we take the `[CLS]` token embedding as the image feature $\mathbf{v} \in \mathbb{R}^{768}$.

Text Encoder. For multilingual text encoding, we adopt mBERT (*bert-base-multilingual-cased*) (Devlin et al., 2019). The OCR-extracted advertisement text (for CN) or original text (for

EN) is tokenized and encoded, and the [CLS] representation $\mathbf{t} \in \mathbb{R}^{768}$ serves as the text feature. Maximum sequence length is set to 30 tokens.

Sentiment Encoder. The sentiment branch encodes auxiliary sentiment information into a 768-dimensional embedding $\mathbf{s} \in \mathbb{R}^{768}$. This embedding is derived from a learned sentiment representation module trained on the MultiMM sentiment labels, providing a complementary affective signal for the metaphor detection task. Concretely, the sentiment module is first trained on MultiMM training pairs using only the sentiment supervision signal, and its weights are subsequently frozen when the metaphor detection head is optimized. This two-stage strategy ensures that the sentiment branch captures genuine affective semantics rather than task-specific artifacts. The embedding dimensionality is set to 768 to match $\mathbf{v}$ and $\mathbf{t}$, enabling symmetric concatenation without additional projection layers.

Fusion and Classification. For metaphor detection, the three feature vectors are concatenated:

$$\mathbf{h} = [\mathbf{v}; \mathbf{t}; \mathbf{s}] \tag{1}$$

For sentiment analysis, only image and text features are concatenated:

$$\mathbf{h} = [\mathbf{v}; \mathbf{t}] \tag{2}$$

The fused representation $\mathbf{h}$ is passed through a feed-forward classifier with dropout (rate 0.3) to produce the final prediction:

$$\hat{y} = \text{softmax}(\mathbf{W}\,\text{dropout}(\mathbf{h}) + \mathbf{b}) \tag{3}$$

We also compare concatenation with element-wise addition and max-pooling fusion strategies in our ablation study (Section 6.5).

5.3 Training Details

Both the ViT and mBERT encoders are initialized from pre-trained weights and kept frozen during training. Only the classification head and sentiment embedding module are fine-tuned. We use the AdamW optimizer (Loshchilov and Hutter, 2018) with a learning rate in the range $[3 \times 10^{-5}, 5 \times 10^{-4}]$, cross-entropy loss, batch size 64, dropout 0.3, and train for 10 epochs. All experiments are implemented in PyTorch (Paszke et al., 2019).

6 Experiments

6.1 Baselines

We evaluate three categories of baselines:

Text-Only Models. mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), SixTP (Chen et al., 2022), LaBSE (Conneau et al., 2020), GPT-3.5-Turbo (Ouyang et al., 2022), Llama2-8B (Touvron et al., 2023), Llama3.1-70B (Dubey et al., 2024), and DeepSeek R1-70B (DeepSeek-AI et al., 2025). These models use only the textual component of each sample.

Vision-Only Models. VGG16 (Simonyan and Zisserman, 2014), ResNet50 (He et al., 2016), and ViT (Dosovitskiy et al., 2020), using only the image modality.

Multimodal Models. mBERT-Res (mBERT + ResNet50 fusion), CMGCN (Liang et al., 2022), CES (Blaier et al., 2021), MET (Xu et al., 2022), MFC (Maity et al., 2022), GPT-4o (Achiam et al., 2023), LLaVA (Liu et al., 2024), Gemini (Team et al., 2023), Qwen2.5-VL-3B/7B/72B (Bai et al., 2025).

6.2 Evaluation Metrics

Following prior work, we report macro-F1 (F1) as the primary metric, along with accuracy and macro-precision. For cross-cultural comparison, we report results separately for CN and EN test sets.

6.3 Main Results

Table 2 reports the main experimental results on both tasks. SEMD achieves the best performance across all settings, with F1 scores of **80.16%** (EN) and **77.79%** (CN) for metaphor detection, and **75.69%** (EN) and **70.51%** (CN) for sentiment analysis.

Text vs. Vision vs. Multimodal. Text-only models achieve up to 69.90% F1 (XLM-R, EN) on metaphor detection, while vision-only models reach up to 71.67% (ViT, EN). Among unimodal models, ViT consistently outperforms text encoders on metaphor detection, highlighting the importance of visual information for this task. Multimodal models substantially outperform unimodal counterparts, confirming that both modalities are necessary for high-quality multimodal metaphor understanding.

MLLMs Underperform. Large multimodal language models—GPT-4o, LLaVA, Gemini, and Qwen2.5-VL variants—systematically underperform specialized multimodal baselines, particularly on Chinese data. GPT-4o achieves only 64.00% (EN) and 67.00% (CN) F1 on metaphor detection,

Model	Metaphor Detection F1 (%)		Sentiment Analysis F1 (%)	
	EN	CN	EN	CN
Random	46.70	41.20	40.41	35.02
mBERT	64.00	65.52	62.83	58.43
XLM_R	69.90	62.38	69.44	66.21
SixTP	68.61	65.60	64.43	65.59
LaBSE	68.15	66.41	69.93	69.02
GPT-3.5-TURBO	30.95	15.49	42.06	52.08
Llama2-8B	22.58	19.73	40.11	50.72
Llama3.1-70B	44.04	49.70	33.46	30.90
Deepseek R1-70B	44.06	47.95	53.38	49.01
VGG16	70.64	67.85	59.90	59.02
ResNet50	71.46	68.56	57.49	58.86
ViT	71.67	69.04	61.46	62.17
mBERT-Res	77.83	74.40	69.75	68.84
CMGCN	79.04	74.91	72.79	70.19
CES	78.45	75.53	73.24	70.26
MET	76.04	72.51	71.95	67.17
MFC	75.84	73.25	71.93	69.60
GPT-4O	64.00	67.00	36.00	29.00
LLaVA	73.31	69.84	56.72	53.21
Gemini	64.19	68.51	53.61	43.48
Qwen2.5-VL-3b	57.30	41.69	52.50	55.26
Qwen2.5-VL-7b	56.62	54.51	50.71	55.15
Qwen2.5-VL-72B	59.12	67.66	50.44	57.17
SEMD	**80.16**	**77.79**	**75.69**	**70.51**

Table 2: Experimental results on metaphor detection and sentiment analysis tasks. The best overall results are highlighted in **bold**, while the best results within each modality are underlined. Models are categorized by modality, with textual, visual, and multimodal models highlighted in cyan, green, and orange, respectively.

while Qwen2.5-VL-72B reaches 59.12% (EN) and 67.66% (CN). This suggests that these models, despite their general-purpose capabilities, have not been sufficiently aligned with the cultural and linguistic nuances required for multimodal metaphor understanding.

Cultural Bias. All model classes exhibit a consistent performance gap between English and Chinese, with Chinese consistently harder. For example, SEMD achieves 80.16% (EN) vs. 77.79% (CN) on metaphor detection. The gap is more pronounced for sentiment analysis (75.69% EN vs. 70.51% CN). This confirms that Chinese multimodal metaphors are inherently more challenging—likely due to subtler visual-linguistic cues, richer contextual dependencies, and greater lexical ambiguity in sentiment expression.

Error Analysis. Metaphor detection accuracy is 86% (EN) vs. 73% (CN), confirming greater difficulty in Chinese. For sentiment analysis, neutral sentiment detection is 62% (EN) vs. 55% (CN), reflecting greater lexical ambiguity in Chinese sentiment expression. Qualitative examination of misclassified samples reveals that false positives most often involve visually complex literal advertisements with unusual color contrasts or object juxtapositions that superficially resemble metaphorical imagery. False negatives are more common for subtle Chinese metaphors in which the source domain is conveyed entirely through image composition, with no explicit textual marker, underscoring the need for stronger object-level visual attention mechanisms and culturally grounded visual representations in future models.

		mBERT	mBERT-Res	SEMD
EN → CN	Acc (%)	64.29	76.35	78.57
	Pre (%)	63.72	76.73	78.19
	F1 (%)	59.70	75.63	77.64
CN → EN	Acc (%)	65.22	73.86	75.45
	Pre (%)	67.46	75.90	77.73
	F1 (%)	65.31	73.96	75.53

Table 3: The results of metaphor detection tasks after Chinese-English translation. Experimental results on metaphor detection task via bidirectional Chinese-English translation.

6.4 Translation Experiments

To directly measure the cultural transfer gap, we conduct cross-lingual translation experiments: English test samples are machine-translated to Chinese (EN→CN) and evaluated on the CN-trained model, and vice versa. Results are reported in Table 3.

SEMD consistently outperforms mBERT and mBERT-Res under both translation directions. However, all models suffer significant performance drops compared to in-language evaluation: SEMD achieves 77.64% F1 on EN→CN vs. 77.79% in-language CN, and 75.53% F1 on CN→EN vs. 80.16% in-language EN. This gap reveals that language translation alone cannot bridge cultural differences in metaphor conventions: the visual and cultural context remains distinct even when the text is translated. Notably, the performance degradation is asymmetric across directions: the EN→CN drop is smaller for metaphor detection (0.15% absolute), whereas the CN→EN direction degrades more substantially (2.52% absolute). This asymmetry suggests that English advertisements rely more heavily

Sentiment Feature	Fusion Method	EN F1 (%)	CN F1 (%)
w/o	add	75.95	72.05
	max	76.29	74.09
	concat	<u>78.64</u>	<u>74.84</u>
w/	add	77.76	74.37
	max	78.59	74.37
	concat	**80.88**	**77.39**

Table 4: Results of the ablation study. The best and second-best results are highlighted in **bold** and <u>underline</u>, respectively.

on textual cues that survive translation, while Chinese advertisements encode metaphorical meaning through tighter text-image co-dependencies that are disrupted when the original text is replaced. These results underline the importance of culturally-aware data collection and modeling. Sentiment analysis proves somewhat more resilient to translation than metaphor detection: SEMD retains a higher proportion of its in-language sentiment F1 under cross-lingual transfer, likely because visual sentiment cues are less language-dependent than fine-grained metaphorical mappings. This finding motivates future work on vision-dominant transfer strategies for cross-cultural multimodal metaphor processing.

6.5 Ablation Study

Table 4 reports the ablation study results, comparing three fusion strategies (add, max-pooling, concatenation) with and without the sentiment feature branch.

Sentiment Feature. Adding the sentiment branch consistently improves performance across all fusion strategies and both languages. The gain is largest for concatenation fusion: from 78.64% to 80.88% F1 (EN) and from 74.84% to 77.39% (CN). This confirms that affective signals provide a complementary and culturally cross-lingual signal for multimodal metaphor detection.

Fusion Strategy. Concatenation outperforms both element-wise addition (add) and max-pooling (max) in all configurations. Without sentiment, concatenation achieves 78.64% (EN) and 74.84% (CN)—already the best among w/o-sentiment variants. With sentiment, concatenation yields the best overall result of 80.88% (EN) and 77.39% (CN). We note that the main results table reports SEMD with the full development-tuned configuration (F1: 80.16% EN, 77.79% CN), which applies additional regularization during final training.

7 Conclusion

We presented MultiMM, the first cross-cultural multimodal metaphor benchmark covering 8,461 Chinese and English advertisement text-image pairs with fine-grained metaphor, domain, and sentiment annotations. Through comprehensive evaluation of 18 baselines, we demonstrated systematic cultural bias in multimodal metaphor processing: all model classes perform worse on Chinese than English, MLLMs underperform specialized models, and translation-based cross-cultural transfer fails to fully bridge the cultural gap. To address these challenges, we proposed SEMD, a sentiment-enriched multimodal fusion model that achieves state-of-the-art performance on both metaphor detection and sentiment analysis across both languages. Our analyses of cultural divergence in sentiment distribution and source domain vocabulary provide concrete empirical evidence for culture-specific metaphor conventions, and translation experiments quantify the remaining gap that purely linguistic adaptation cannot close.

Limitations. MultiMM is currently limited to Chinese and English advertisements and may not generalize to other language pairs or content domains such as social media or news imagery. The sentiment annotations may themselves carry implicit cultural assumptions of the annotator pool despite multicultural recruitment efforts. Additionally, SEMD freezes its pre-trained visual and textual encoders during training, which limits end-to-end optimization; joint fine-tuning with carefully controlled, modality-specific learning rates is a promising direction for future work.

Our work opens several avenues for future research. Richer cultural knowledge, including region-specific commonsense and visual symbol inventories, could be incorporated into multimodal metaphor models. Extending MultiMM to additional language pairs and advertisement genres would further characterize the scope of cultural bias. Finally, the strong relationship between sentiment and metaphoricity, evidenced here, suggests that multitask learning approaches jointly modeling affect and figurative language could yield further gains in both cross-cultural robustness and interpretability.

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